

Abhirup Saha

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🐙 GitHub profile

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Education

Indian Institute of Engineering Science and Technology, Shibpur

B.Tech in Computer Science and Technology; CGPA: 9.56/10 (upto 6th Semester)

Aug, 2023 – June, 2027

Howrah, WB, India

Kendriya Vidyalaya No.2, Saltlake

CBSE; AISSCE (12th) Score: 95%

April, 2023

Saltlake, WB, India

Kendriya Vidyalaya No.2, Saltlake

CBSE; AISSE (10th) Score: 96.4%

April, 2021

Saltlake, WB, India

Work Experience

RESEARCH INTERN at VECC — Nuclear Pulse Denoising 🔗

May, 2026 – July, 2026

- Architected a 4-stage physics-guided **Variational Autoencoder (VAE-PGNN)** pipeline to denoise nuclear pulse signals, achieving a **10× baseline noise reduction** and **+22 dB SNR improvement** on experimental radioisotope data.
- Synthesized a large-scale training dataset via **Monte Carlo domain randomization**, and benchmarked the VAE against classical filters (**Wiener, Kalman**) optimized using various **meta-heuristic** algorithms.

Projects

SMB Client App: Android app to access SMB file-shares 🔗

February, 2026 – August, 2026

- Engineered a **Flutter (Dart)** Android application for accessing **SMBv2/3** network shares, implementing NTLM authentication and hierarchical file navigation for **10,000+** files.
- Secured user credentials with hardware-backed **AES-256 GCM** encryption via Android Keystore and mitigated MITM risks through strict local subnet IP validation with **0%** data leakage.
- Optimized file operations by building an asynchronous multi-file download manager with in-memory **ZIP compression**, reducing storage footprint by **30%** and cutting download times by **25%**.

Notely: Microservices-based Note-taking Platform 🔗

April, 2026 – August, 2026

- Designed a robust **microservices architecture** using an **Express API gateway**, supporting up to **5,000** concurrent users with independent authentication, user, and note services.
- Containerized the entire backend infrastructure using **Docker** and **Docker Compose**, improving deployment speed by **50%** with polyglot persistence utilizing **PostgreSQL, MongoDB, and Redis**.
- Developed a responsive Single-Page Application (SPA) using **React**, served via **Nginx**, achieving a **99.9%** uptime and **<200ms** API response times.

Beyond Rules: Adaptive Fraud Detection for Banks 🔗

June, 2025 – September, 2025

- Developed a **multi-layered AI fusion** system to detect banking fraud, analyzing over **10,000** real-time transactions per second using **FastAPI** and **PyTorch**.
- Integrated an on-device **TFLite** behavioral model and a **Graph Neural Network (GNN)** utilizing **Neo4j** to perform dual-validation, reducing false positives by **40%**.
- Built a cross-platform **Flutter** application for banking clients, decreasing fraud detection latency by **35%** while detecting complex anomalous patterns.

IPL Data Analysis and Winner Prediction 🔗

March, 2025

- Engineered a predictive ML pipeline leveraging a **StackingClassifier** (RandomForest & SVC), fine-tuned via **hyperopt** to forecast IPL match winners with **79.71% accuracy** and **79.99% F1-score**.
- Conducted comprehensive **EDA** and **feature engineering** on **17 seasons** of IPL data, resolving data inconsistencies and deriving **5 new analytical datasets** to map team and player performance trends.
- Designed and evaluated an **Artificial Neural Network (ANN)** using **PyTorch** with **SiLU** activations alongside classical ensembles, demonstrating the superiority of ensemble methods on smaller tabular datasets.

- Engineered a scalable **web scraper** leveraging **Selenium**, **BeautifulSoup**, and **parallel processing** to compile and sanitize a curated dataset of over **1,300** Indian news articles.
- Fine-tuned a **BigBird-RoBERTa** transformer model to classify long-form political text into ideological leanings, optimizing a weighted **F-beta score** (1.8) to counteract severe class imbalance.
- Deployed a full-stack **FastAPI** web application to serve real-time predictions via URL or raw text input, integrating **Hugging Face** tokenizers and **PyTorch** inference engines.

Achievements

- Ranked **1st** among 29 teams in **Truth Tuner, 2025**, building an **ML model** for Indian News Bias Detection
- **Finalist** in **IIESTxUCO Bank Hackathon** on **Fraud Detection in Banking Transactions**

Technical Skills

Languages: C, C++, Python, Dart, JavaScript, HTML, CSS, SQL

Frameworks and Libraries: Flutter, React, Selenium, NumPy, Pandas, Scikit-Learn, Matplotlib

Software Tools: Docker, Git, GitHub, Android Studio, VS Code, Linux

Core Subjects: OOPS, CN, OS, DBMS

Team Experience

LEAD . CORE MEMBER — Cybersecurity Chapter : CodeIIEST, GDGoC IIESTS October 2025 – Present

- Successfully organized **RootAccess CTF**, **EOFool CTF** in AY 2025-26 with **300+** participants.
- Held multiple events on concepts of **Cryptography**, **Computer Networks**.